

Multimedia in interconnected LAN systems

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Bridges and routers that are used to interconnect today's local area networks (LANs) are likely to limit the range of multimedia applications that can be supported on such networks. The key problem is end-to-end delay and jitter (delay variation). This paper quantifies these delays.

1. Introduction

Networked multimedia applications can be delivered to the desk today via PBXs or LANs, but both systems have their limitations. PBX solutions can provide low end-to-end delay but only at relatively low bandwidth, whereas LANs can deliver high bandwidth, but the delay is unpredictable. The importance of these limitations depends on the relative significance of the real time (audio and video) and non-real time (data) components that are used by the multimedia applications. This paper examines the delays that can be encountered in interconnected LAN systems and possible ways of constraining them.

2. Multimedia application categories

Multimedia applications can be divided into two broad categories, namely 'interactive' and 'retrieval'. Both types can include any mix of audio, video and data but the interactive category is much more sensitive to delay and delay variations (jitter). Video telephony clearly comes in the interactive category and human factors dictate that the end-to-end delay must be kept to a minimum. Other interactive applications that are sometimes associated with video telephony are shared wordprocessors, spreadsheets or drawing packages.

Retrieval applications such as distance learning are more delay-tolerant since users will readily accept a delay of a few seconds between requesting information and having the information presented on their screens. Retrieval applications can also benefit from the use of asymmetric encode/decode algorithms which use a simple, low-delay, decode algorithm at the expense of a more complex encoding algorithm. This is acceptable for retrieval applications since the encoding is only done once and does not have to be done in real time.

Delay variation is more problematical than absolute delay for both categories of applications, but here again

retrieval applications have an advantage since buffers can be introduced to smooth out any variations and thereby minimise the impact on the end user. The following discussion applies equally to all types of applications but the use of retrieval applications may grow faster in the short term because they are less demanding on the technology.

3. Local area networks

Traditional LANs such as Ethernet[™] were originally designed to interconnect a number of computers and peripherals across a shared, passive co-axial bus. Attached nodes check that no other nodes are using the bus before they transmit each packet of data. This access protocol can obviously introduce significant delay variations entirely dependent on the volume of traffic from other nodes attached to the network. The problem can be constrained by limiting the number of nodes connected to each network and today, with structured wiring where the co-axial bus is replaced with twisted pairs, it is common to have one node per network.

The hubs used to interconnect such networks do so with multiport bridges and/or routers [1] which use store and forward technology. For data applications this is fine, since it enables a high level of error detection and correction (by re-transmission) to be introduced, but the delay variations that can also be introduced are likely to cause havoc to any real time elements of multimedia applications. In the short term a number of users will tolerate the delay problems, especially on lightly loaded networks. However, most users are likely to wait for the next generation of routers that will incorporate the new IETF standards which specifically address this problem. The first such routers are expected to become available in late 1996.

The following description, together with Fig 1, illustrates how the main components of a LAN-based

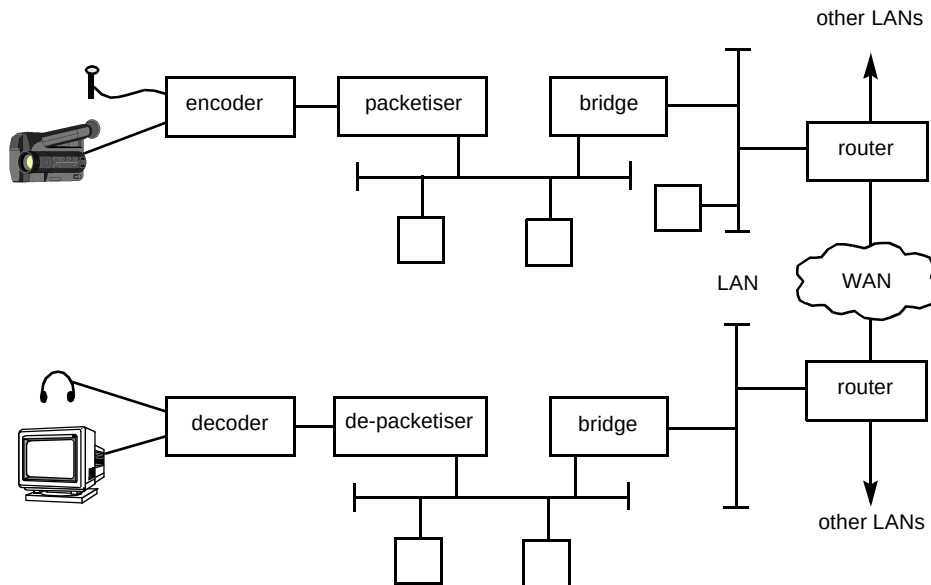


Fig 1 Typical extended local area network.

network contribute to the overall end-to-end delay as seen by an interactive application. Additional unlabelled boxes are shown in Fig 1 to emphasise the fact that LANs are shared. These boxes could be personal computers, workstations, print servers or bridges and routers linking to other networks.

The encoder converts the analogue signal into a digital data stream. The delay is dependent on the coding scheme and compression algorithm used. Algorithms that provide high compression ratios also introduce higher delays. For H.261 encoded video at 128 kbit/s the delay is approximately 180 ms. The delay through the packetiser depends on the packet size and for Ethernet systems this can vary from 64 bytes to 1518 bytes (including overheads), with corresponding delays of between 4 ms and 95 ms. This presents a classic 'catch 22' problem in that to keep the end-to-end delay down, small packets should be used, but due to the fixed per-packet overheads small packets increase the overall level of network traffic. This delay is entirely deterministic and Table 1 illustrates how the delay varies with packet size and clock rate. The delay across the physical LAN is measured in microseconds and can generally be ignored.

Table 1 Serialisation delay (ms).

Clock rate	Packet size (bytes)			
	64	256	1000	1518
64 kbit/s	8	32	125	190
128 kbit/s	4	16	63	95
384 kbit/s	1	5	21	32
1500 kbit/s	0.3	1	5	8
2048 kbit/s	0.3	1	4	6

Bridges typically extend LANs within a building or campus site and although Fig 1 shows a two-port bridge it is

common to have multiport units capable of switching the traffic between any ports. Bridges read in the complete data packet and check it for errors before forwarding it. Assuming the data is error free the bridge accesses the destination network in exactly the same way as any other device and must wait until the network is idle before transmitting the data. Data that cannot be immediately transmitted is temporarily stored in a FIFO (first-in-first-out store) and this is where the delays can build up depending on the level of network traffic. Under worst-case conditions bridges can introduce delay variations of a few hundred milliseconds although a few tens of milliseconds is more typical.

Routers are also multiport devices and are typically used to interconnect LANs across wide area networks although they can be used in place of bridges to give more control over the traffic flows. Routers operate in a similar manner to bridges, but the delay variations can be much more significant, particularly when used to connect across wide area networks [2]. This is due to the fact that the wide area network bandwidth is typically only a fraction of the LAN bandwidth and the delay through the FIFO varies as the traffic bursts come and go. Table 2 illustrates the delay that can build up in a router depending on the size of the FIFO and the line rate of the wide area network interface.

Table 2 Buffer capacity (seconds).

Clock rate	Memory size (kbytes)		
	128	512	1024
64 kbit/s	16.0	64.0	128.0
128 kbit/s	8.0	32.0	64.0
384 kbit/s	2.7	10.7	21.3
1500 kbit/s	0.7	2.7	5.5
2048 kbit/s	0.5	2.0	4.0

The delay across the wide area network is approximately 8 μ s/mile, hence if the link is short this delay can be ignored, but not if the link goes around the world or involves a satellite hop.

The depacketiser introduces similar delay to the packetiser and the video decoder introduces approximately 120 ms assuming 128-kbit/s H.261 video.

It can be seen from the above figures that routers and to a lesser extent, bridges are the main source of delay variation and hence represent the key problem areas in supporting interactive multimedia applications on distributed LANs. Figure 1 shows a single bridge and router at each end of the extended network, but real systems are more likely to have a number of these units in series and hence the delay variation problems become even worse.

Traffic profiles on today's LANs vary widely but most users would not notice if a few multimedia demonstrators were added to existing networks. However, large-scale roll out of such applications would require network upgrades. LAN vendors are coming up with a number of schemes to reduce the problems and extend the life of the installed systems, but to date such solutions only work in specific cases and are proprietary in nature. A new standard that directly addresses the delay problem is expected to be ratified during 1996. This standard introduces new, quality of service information into the packet headers so that the intermediate systems (routers) can identify the different types of data and expedite the forwarding of real time data at the expense of other traffic.

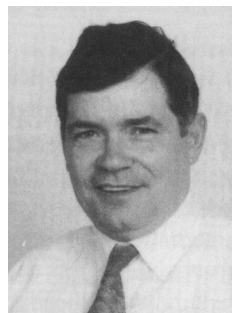
Another solution is isochronous Ethernet (IEEE 802.9a). This multiplexes packet and circuit-switched traffic over the same physical wire. Today this offers the best technical solution for interactive multimedia applications, since the real time interactive traffic can go over the low-delay connection-oriented network and the data traffic over the connectionless network. Some switch vendors are adding this as an optional interface card in their PBXs but it remains to be seen whether the market takes this up.

4. Conclusions

If the packet network world gets its new resource reservation standard correct, then future multimedia applications that include audio and video will be delivered to the desk via LANs. PBXs will remain dominant for low-bandwidth applications such as voice only, but there is little if any growth in this area and the LAN could displace rather than work along side the PBX in many businesses.

References

- 1 Pearlman R: 'Interconnections (Bridges and Routers)', Addison-Wesley (1992).
- 2 Miller M: 'Internetworking (LAN to LAN, LAN to WAN)', Prentice Hall (1991).



Bill Bunn has worked on computer networking and related aspects for the past 12 years and currently has responsibility for co-ordinating the top level design of BTnet and evaluating evolution options.

Of particular interest is how migration to IPng can be achieved, with minimum operational pain, and the impact new IETF protocols such as RTP and RSVP will have on BT's competitive position with respect to transporting real time traffic.